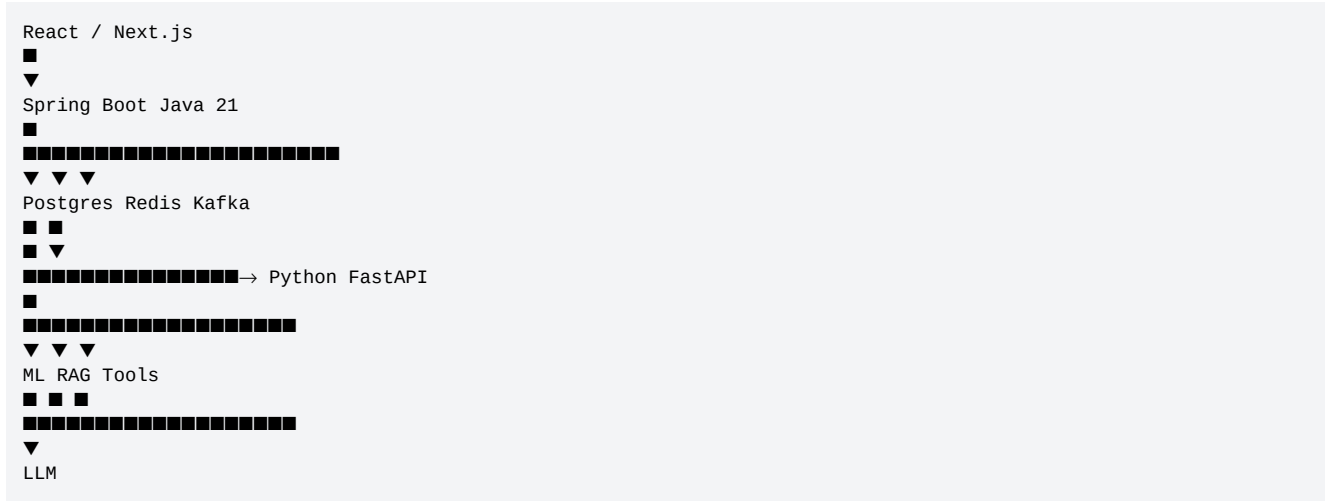


High-Level Design

ServiceGuard AI

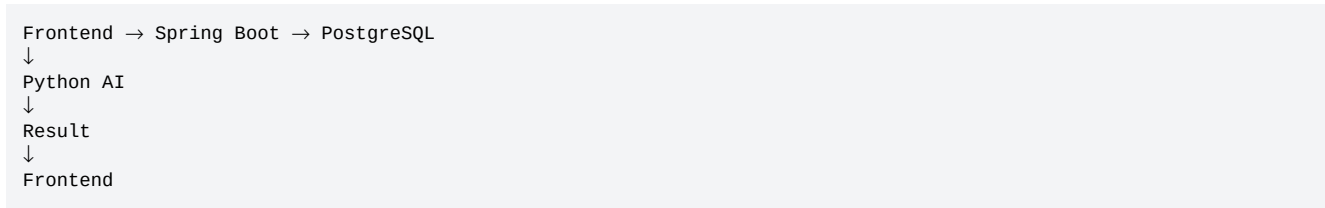
1. Architecture



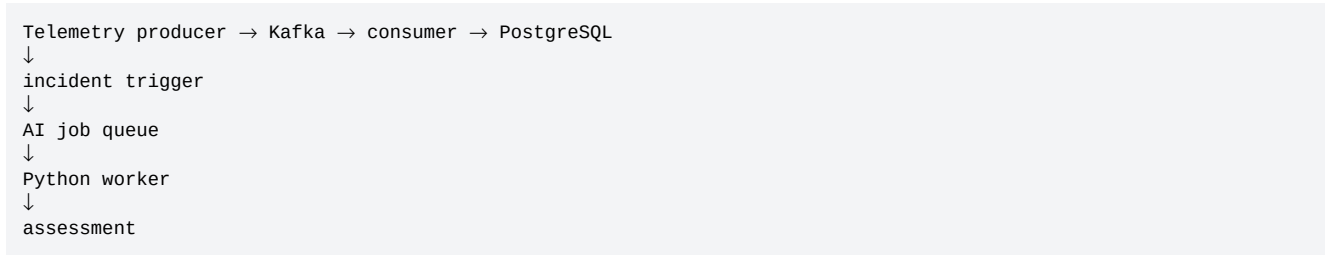
2. Service responsibilities

Service	Owns
Spring Boot	Auth, RBAC, business rules, CRUD, incident lifecycle, audit, API contracts
PostgreSQL	Transactional source of truth, semantic views, pgvector
Redis	Rate limiting, cache, short-lived state
Kafka	Telemetry/event streaming and async jobs when required
Python FastAPI	ML inference, RAG, agent orchestration, external integrations
Frontend	Control tower UI and AI copilot

3. Synchronous path



4. Asynchronous path



5. Architecture principle

Keep the MVP as a modular system first. Add Kafka, distributed workers and additional infrastructure only when the workload or reliability requirements justify them.